Rectangular matrix multiplication from polynomial products

Przemysław Uznański

Pathway przemek@pathway.com

How far can the methods behind the bound $\omega \leq 2.25$ be pushed for rectangular products?

Preliminary slides ▪ October 2026

Based on the accompanying note. Slides prepared with GPT-6.1.

What is ω?

Given $A, B \in \mathbb{C}^{n \times n}$, compute $C = AB$ exactly.

The matrix-multiplication exponent

ω is the infimum of exponents τ for which some algorithm multiplies two $n \times n$ matrices using $O(n^\tau)$ field operations.

The usual algorithm

n^2 dot products of length n:

$$O(n^3), \qquad \omega \leq 3.$$

Strassen's algorithm

Seven recursive products of half-size blocks:

$$O(n^{\log_2 7}), \qquad \omega \leq \log_2 7 \approx 2.80735.$$

There are n^2 outputs, so $\omega \geq 2$. The central open question is whether $\omega = 2$.

We count field operations over $\mathbb{C}$, as in the note. Throughout, $O(n^e)$ suppresses a $+o(1)$ term in the exponent. Strassen: Gaussian elimination is not optimal (1969).

A short history of square matrix multiplication

Year	Algorithm / analysis	Upper bound on ω
Classical	Entry-by-entry dot products	3
1969	Strassen	$\log_2 7 \approx 2.80735$
1990	Coppersmith–Winograd	2.375477
2014	Le Gall	2.3728639
2024	Alman et al.	2.371339
Aug. 2026	Dupont et al.: optimization + AlphaEvolve	2.371177
Sept. 2026	Polynomial-product / completion analysis	2.258
Oct. 2026	Polynomial-product bound [1]	2.25

This talk asks how the methods behind the 2.25 bound extend to rectangular products.

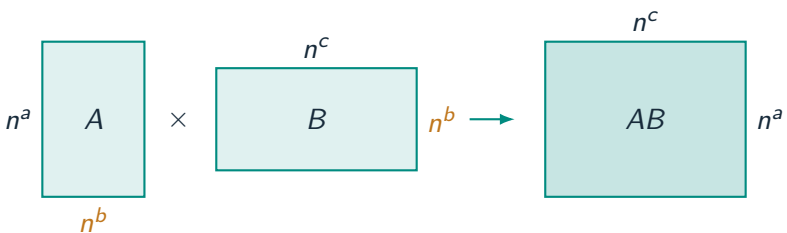
Selected milestones, not an exhaustive timeline. Names link to primary sources; the 2024 entry dates the preprint (published at SODA 2025). The last two entries are the September 24 and October 2 preprints by OpenAI.

What is $\omega(a, b, c)$?

For fixed $a, b, c \geq 0$, consider the product

$$(n^a \times n^b) \cdot (n^b \times n^c) \quad \longrightarrow \quad (n^a \times n^c).$$

Its infimum running-time exponent is $\boldsymbol{\omega(a, b, c)}$.



Exact multiplication over $\mathbb{C}$; dimensions are rounded upward. The shared inner dimension n^b is highlighted. Shapes are schematic.

The rectangular curve and the dual exponent

Two useful special parameters

$$\omega(k) = \omega(1, k, 1), \qquad \omega(1) = \omega.$$

$$\alpha = \sup\{k \geq 0 : \omega(k) = 2\}.$$

The **dual exponent** α measures how far quadratic-time multiplication reaches as the inner dimension grows.

Rectangular milestones

Year	Source	Bound on α
1997	Coppersmith	> 0.29462
2012	Le Gall	> 0.30298
2018	Le Gall–Urrutia	> 0.31389
2024	Vassilevska Williams et al.	≥ 0.321334
Sept. 2026	OpenAI preprint	> 0.465
This note	**Rectangular extension**	$\mathbf{\geq 0.5}$

Motivation: extend the polynomial-product analysis in [1] to the shapes used by all-pairs algorithms, and make the proof accessible to TCS.

These works also bound the curve $\omega(k)$ beyond the quadratic range; α summarizes where that range ends. The 1997–2024 dates are publication years. Dimensions are rounded upward.

A bound for every matrix shape

Sort the three exponents: $a \geq b \geq c \geq 0$.

Theorem 1.1 of the note

$$\omega(a,b,c) \leq \begin{cases} a+b, & c \leq \dfrac{ab}{a+b}, \\ \dfrac{a+b+c}{2} + \dfrac{1}{4}\left(\dfrac{ab}{c} + \dfrac{ac}{b} + \dfrac{bc}{a}\right), & c > \dfrac{ab}{a+b}. \end{cases}$$

If $c \leq \frac{ab}{a+b}$, the exponent is exactly $a+b$.

Permuting tensor legs lets us use the sorted dimensions of any product.

The threshold is interpreted as zero at $a = b = 0$. The largest input size supplies the lower bound $a+b$.

The rectangular curve

Corollary 1.2 of the note

$$\omega(k) \le \begin{cases} 2, & 0 \le k \le \frac{1}{2}, \\ 1 + k + \dfrac{1}{4k}, & k \ge \frac{1}{2}. \end{cases}$$

The thin range is exact.

$\omega(k) = 2$ for $0 \le k \le \frac{1}{2}$.

The dual exponent satisfies $\alpha \ge \frac{1}{2}$.

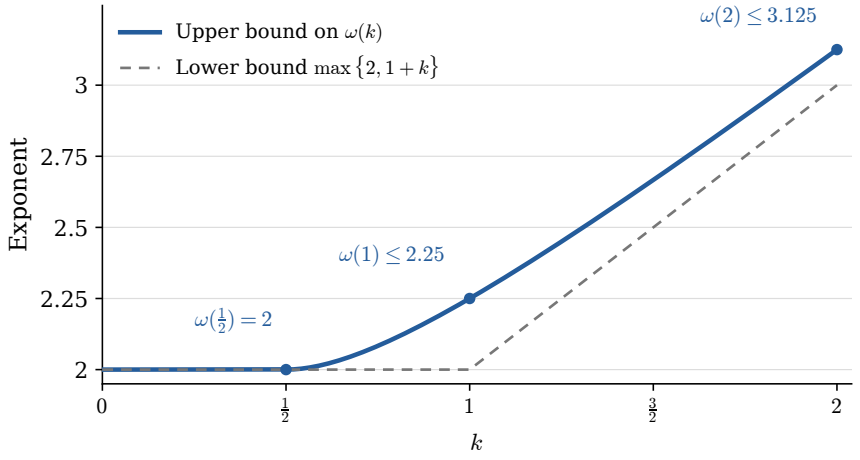
Selected values.

k	1	2
$\omega(k) \le$	2.25	3.125

For $k \ge 1$, the gap above the size lower bound is at most $\frac{1}{4k}$.

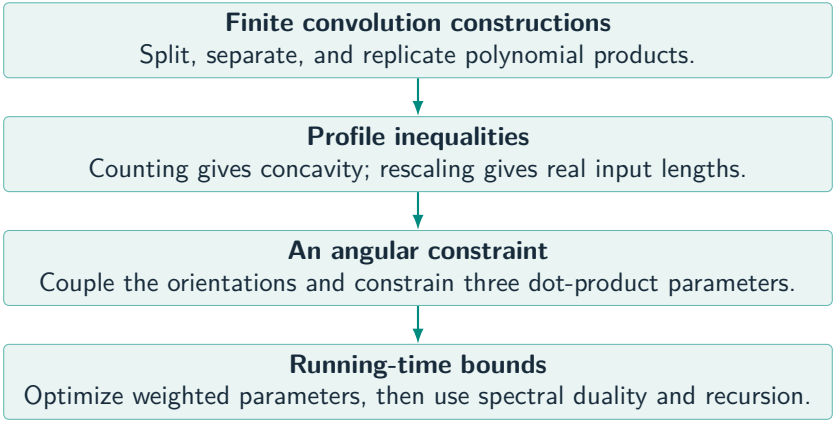
$\alpha = \sup\{k \ge 0 : \omega(k) = 2\}$. The lower bound is $\omega(k) \ge \max\{2, 1 + k\}$.

The bound for $0 \leq k \leq 2$



For $0 \leq k \leq \frac{1}{2}$, the upper bound meets the size lower bound.

The proof in four steps



Sections 2–6 of the note. The spectral theorem is a published input; the rectangular parameter analysis is the extension studied here.

Ordinary block multiplication uses eight products

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, \qquad C = AB.$$

Four outputs, two products each

$$\begin{aligned} c_{11} &= a_{11}b_{11} + a_{12}b_{21}, \\ c_{12} &= a_{11}b_{12} + a_{12}b_{22}, \\ c_{21} &= a_{21}b_{11} + a_{22}b_{21}, \\ c_{22} &= a_{21}b_{12} + a_{22}b_{22}. \end{aligned}$$

We may use linear combinations

1. Add or subtract blocks within each input.
2. Multiply one combination from each input.
3. Add or subtract the products to form the outputs.

Can linear combinations reduce the eight multiplications?

The entries a_{ij} and b_{ij} may be scalars or matrix blocks. Ordinary block recursion gives $T(n) = 8T(n/2) + O(n^2)$, including all additions.

Strassen's seven products and four outputs

For the same matrices A and B, compute $C = AB$ as follows.

Compute these seven products

$$
\begin{aligned}
M_1 &= (a_{11} + a_{22})(b_{11} + b_{22}),\\
M_2 &= (a_{21} + a_{22})b_{11},\\
M_3 &= a_{11}(b_{12} - b_{22}),\\
M_4 &= a_{22}(b_{21} - b_{11}),\\
M_5 &= (a_{11} + a_{12})b_{22},\\
M_6 &= (a_{21} - a_{11})(b_{11} + b_{12}),\\
M_7 &= (a_{12} - a_{22})(b_{21} + b_{22}).
\end{aligned}
$$

Add and subtract to get the outputs

$$
\begin{aligned}
c_{11} &= M_1 + M_4 - M_5 + M_7,\\
c_{12} &= M_3 + M_5,\\
c_{21} &= M_2 + M_4,\\
c_{22} &= M_1 - M_2 + M_3 + M_6.
\end{aligned}
$$

For example, the first combination simplifies to $c_{11} = a_{11}b_{11} + a_{12}b_{21}$.

Seven block products give $T(n) = 7T(n/2) + O(n^2)$, hence $\omega \leq \log_2 7$.

Strassen (1969): Gaussian elimination is not optimal. Every product has a left-input factor followed by a right-input factor; the same formulas therefore apply to matrix blocks.

Tensor rank counts bilinear products

Use formal labels c_{ij} for the four output positions:

$$T = \underbrace{\begin{aligned} &a_{11}b_{11}c_{11} + a_{12}b_{21}c_{11} + a_{11}b_{12}c_{12} + a_{12}b_{22}c_{12} \\ &+ a_{21}b_{11}c_{21} + a_{22}b_{21}c_{21} + a_{21}b_{12}c_{22} + a_{22}b_{22}c_{22} \end{aligned}}_{\text{8 ordinary terms}}$$

$$\begin{aligned} = {} & (a_{11} + a_{22})(b_{11} + b_{22})(c_{11} + c_{22}) \\ & + (a_{21} + a_{22})b_{11}(c_{21} - c_{22}) + a_{11}(b_{12} - b_{22})(c_{12} + c_{22}) \\ & + a_{22}(b_{21} - b_{11})(c_{11} + c_{21}) + (a_{11} + a_{12})b_{22}(c_{12} - c_{11}) \\ & + (a_{21} - a_{11})(b_{11} + b_{12})c_{22} + (a_{12} - a_{22})(b_{21} + b_{22})c_{11}. \end{aligned}$$

7 Strassen terms: one linear form from each **leg** a, b, c.

Tensor rank is the fewest such terms in a sum for T. This identity gives $\operatorname{rank}(\langle 2,2,2\rangle) \leq 7$.

Section 2. Collect each M_r's coefficients in the four output formulas to obtain its linear form in c. For instance, M_2 multiplies $c_{21} - c_{22}$.

Matrix multiplication tensors and the exponent

All earlier improvements to ω fit this **Strassen-type bilinear framework**, including tensor powers and degenerations.

$$\langle m, r, p \rangle = \sum_{i=1}^{m} \sum_{k=1}^{r} \sum_{j=1}^{p} a_{ik} b_{kj} c_{ij}.$$

It encodes $m \times r$ by $r \times p$ multiplication; c_{ij} labels output position (i, j).

A rank-R identity gives an algorithm using R bilinear products. Applying a square identity with fixed block size q recursively gives

$$T(n) \leq R\, T(n/q) + O(n^2).$$

$$\operatorname{rank}(\langle q, q, q \rangle) \leq R \quad \Longrightarrow \quad \omega \leq \log_q R.$$

Historical viewpoint: Alman (2021), Section 1; tensor/rank framework: [3] and Sections 2, 6 of the note. Dimensions are positive integers; $q \geq 2$ is fixed. Linear-map costs are included; the exponent convention absorbs logarithmic factors.

Restriction: reuse an algorithm after linear substitutions

Make linear substitutions independently on the a-, b-, and c-legs.

What happens to the algorithm?

Linearly encode the new inputs, run the old bilinear products, then linearly combine their outputs.

Each product of three linear forms remains a product of three linear forms after substitution.

Example: keep only the first row

Set $a_{21} = a_{22} = 0$ in 2×2 multiplication:

$$\begin{pmatrix} a_{11} & a_{12} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ 0 & 0 \end{pmatrix}.$$

This computes a 1×2 row times a 2×2 matrix:

$$\langle 2, 2, 2 \rangle \geq \langle 1, 2, 2 \rangle.$$

$T \geq S \implies \operatorname{rank}(S) \leq \operatorname{rank}(T)$.
The same bilinear products suffice after the substitutions.

Section 2. The notation $T \geq S$ means that T restricts to S. Substitutions on the formal output labels specify the linear decoding of outputs. Here the zero second row lets us retain only c_{11}, c_{12}.

Direct sum: run independent computations side by side

Use disjoint variables on all three legs:

$$T(a, b, c) \oplus U(a', b', c') = T(a, b, c) + U(a', b', c').$$

Example: two scalar products

$$a_1 b_1 c_1 + a_2 b_2 c_2$$

encodes the two separate outputs

$$c_1 = a_1 b_1, \qquad c_2 = a_2 b_2.$$

How the algorithms combine

If T uses R products and U uses S products, concatenate the lists: $R + S$ products compute both.

We write mT for m independent copies of T, each with its own inputs and outputs.

$$\operatorname{rank}(T \oplus U) \leq \operatorname{rank}(T) + \operatorname{rank}(U).$$

Section 2. Every term from either rank decomposition is still a rank-one term in the combined variable spaces. In the example, c_1, c_2 are formal output labels; their coefficients are the two scalar products.

Why tensor products give block multiplication

Take $q \times q$ blocks, each an $s \times s$ matrix.

Outer	Inner	Paired variable
a_{ik}	a'_{uw}	$a_{(i,u),(k,w)}$
b_{kj}	b'_{wv}	$b_{(k,w),(j,v)}$
c_{ij}	c'_{uv}	$c_{(i,u),(j,v)}$

(i, u) has qs possible values.

The coefficient of $c_{(i,u),(j,v)}$ is

$$\sum_{k=1}^{q} \sum_{w=1}^{s} a_{(i,u),(k,w)}\, b_{(k,w),(j,v)}.$$

This is the usual row–column dot product over all qs inner indices (k, w).

$\langle q, q, q\rangle \otimes \langle s, s, s\rangle \cong \langle qs, qs, qs\rangle$.
Iterating gives $\langle q, q, q\rangle^{\otimes N} \cong \langle q^N, q^N, q^N\rangle$.

Pairing rank-one terms also gives $\operatorname{rank}(T \otimes S) \le \operatorname{rank}(T)\operatorname{rank}(S)$.

Paired variables are independent coordinates. Relabel $(i, u) \leftrightarrow (i-1)s + u$, and similarly for the column and inner indices. Expanding the two tensors gives the displayed output coefficient; q, s, N are positive integers.

Degeneration: a limit of restrictions

Fix tensors T and S: T is the available computation; S is the desired computation.

Definition

A **degeneration** $T \trianglerighteq S$ is a family of linear substitutions on the three legs of T, depending on $\varepsilon \neq 0$, whose resulting tensors $P(\varepsilon)$ converge **coefficient by coefficient** to S as $\varepsilon \to 0$.

$$T \geq P(\varepsilon) \quad (\varepsilon \neq 0), \qquad \lim_{\varepsilon \to 0} P(\varepsilon) = S.$$

For every $a_i b_j c_k$, its coefficient in $P(\varepsilon)$ must tend to its coefficient in S.

Definition: Section 2 of the note. Substitutions act independently on the three legs; all $P(\varepsilon)$ use the same coordinates as S.

A degeneration that cannot be a restriction

Available: $T = a_0' b_0' c_0' + a_1' b_1' c_1'$ (two independent scalar products).

Target: $S = a_0 b_0 c_0 + (a_0 b_1 + a_1 b_0) c_1$.

For $\varepsilon \neq 0$, substitute independently on the three legs:

$$\begin{aligned} &a_0' \mapsto a_0, && b_0' \mapsto b_0, && c_0' \mapsto c_0 - c_1/\varepsilon, \\ &a_1' \mapsto a_0 + \varepsilon a_1, && b_1' \mapsto b_0 + \varepsilon b_1, && c_1' \mapsto c_1/\varepsilon. \end{aligned}$$

$$\begin{aligned} P(\varepsilon) &= a_0 b_0 \left(c_0 - \frac{c_1}{\varepsilon} \right) + (a_0 + \varepsilon a_1)(b_0 + \varepsilon b_1) \frac{c_1}{\varepsilon} \\ &= S + \varepsilon a_1 b_1 c_1. \end{aligned}$$

The substitutions contain $1/\varepsilon$ and are undefined at zero, but their divergent terms cancel: $P(\varepsilon) \to S$.
$T \trianglerighteq S$, whereas $T \not\geq S$: $\operatorname{rank}(T) = 2 < 3 = \operatorname{rank}(S)$.

All primed variables are independent. The target is the standard W-tensor with its output coordinates exchanged; see [6, proof of Proposition 13, p. 131]. Rank-three proof in backup.

Asymptotic rank is enough to bound ω

The **asymptotic rank** measures the growth of rank under tensor powers:

$$\widetilde{R}(T) := \lim_{N\to\infty} \operatorname{rank}(T^{\otimes N})^{1/N} = \inf_{N\geq 1} \operatorname{rank}(T^{\otimes N})^{1/N}.$$

Fix an integer $q \geq 2$ and put $M = \langle q, q, q\rangle$. A rank decomposition of $M^{\otimes N} \cong \langle q^N, q^N, q^N\rangle$ gives a block algorithm of side q^N. Recursing gives

$$\omega \leq \log_{q^N} \operatorname{rank}(M^{\otimes N}) = \frac{\log_q \operatorname{rank}(M^{\otimes N})}{N}.$$

Taking the infimum over N gives the same conclusion as a rank bound:

$$\widetilde{R}(\langle q, q, q\rangle) \leq R \quad \Longrightarrow \quad \omega \leq \log_q R.$$

Definition: Section 3; algorithm conversion: Section 6. Submultiplicativity of rank ensures the limit exists. Each tensor power and its finite rank identity are fixed before the input-size limit; linear-map costs are included.

Interpolation makes the asymptotic computation exact

Let $P(\varepsilon) = S + \varepsilon P_1 + \cdots + \varepsilon^L P_L$ describe a degeneration $T \trianglerighteq S$. Every nonzero evaluation is a restriction of T.

Take N tensor factors: $P(\varepsilon)^{\otimes N}$ has degree at most NL.

↓

Interpolate at $NL + 1$ nonzero points to recover the constant coefficient $S^{\otimes N}$.

$$\operatorname{rank}(S^{\otimes N}) \leq (NL+1)\operatorname{rank}(T)^N.$$

Take N-th roots: $(NL+1)^{1/N} \to 1$. Thus $\widetilde{R}(S) \leq \operatorname{rank}(T)$.

Lemma 3.1 and Appendix D; standard interpolation conversion [6]. Interpolation is an exact linear combination of evaluations. For matrix multiplication, a fixed tensor power supplies a block algorithm to recurse.

Tensor characters turn constructions into inequalities

A **tensor character** λ assigns a nonnegative real number $\lambda(T)$ to each tensor T, with these four rules:

Independent computations	$\lambda(S \oplus T) = \lambda(S) + \lambda(T)$
Tensor products	$\lambda(S \otimes T) = \lambda(S)\lambda(T)$
One scalar product	$\lambda(abc) = 1$
Restriction	$T \geq S \implies \lambda(T) \geq \lambda(S)$

Every character gives a lower bound on asymptotic rank: $\lambda(T) \leq \widetilde{R}(T)$.

Section 3. The rules give $\lambda(T) \leq \operatorname{rank}(T)$, since $\operatorname{rank}(T)$ independent scalar products restrict to T. Applying this to powers gives $\lambda(T)^N = \lambda(T^{\otimes N}) \leq \operatorname{rank}(T^{\otimes N})$; take N-th roots and the limit.

Strassen's spectral theorem

For every nonzero tensor T over $\mathbb{C}$:

Spectral duality

$$\widetilde{R}(T) = \max_{\lambda} \lambda(T),$$

where the maximum ranges over all tensor characters.

A bound $\lambda(T) \leq R$ for **every** character implies $\widetilde{R}(T) \leq R$.

For $M = \langle q, q, q \rangle$, with integer $q \geq 2$,

$$\lambda(M) \leq R \text{ for every } \lambda \quad \Longrightarrow \quad \omega \leq \log_q R.$$

Theorem 3.2 of the note; Strassen [2, Theorem 3.8] and Wigderson–Zuiddam [3, Corollary 3.30, author version]. The supremum is attained. The exponent implication uses the fixed-power block algorithm from slide 19.

Convolution and dot-product tensors

Write $x^{(X)} = x$, $x^{(Y)} = y$, $x^{(Z)} = z$ for the three groups of variables.

Convolution

$$C_{ij}(r,s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_\ell^{(j)} x_{h+\ell}^{(k)}.$$

Encodes polynomial multiplication. Inputs have r, s coefficients; the output has $r + s - 1$ coefficients.

Oriented dot products

$$B_X(q) = x \sum_{h=1}^{q} y_h z_h,$$

$$B_Y(q) = y \sum_{h=1}^{q} x_h z_h,$$

$$B_Z(q) = z \sum_{h=1}^{q} x_h y_h.$$

C_{ij}: input legs i, j; output leg k.
B_i: leg i is scalar; the other two legs each have q coordinates.

Definitions: Section 2 of the note; convolution background [5]. The legs i, j, k are distinct elements of $\{X, Y, Z\}$; r, s, q are positive integers.

Three parameters describe dot-product values

Fix an arbitrary tensor character λ. For the dot-product tensor $B_X(q) = x \sum_{j=1}^{q} y_j z_j$, and the other two orientations, pairing indices gives

$$B_i(q) \otimes B_i(r) \cong B_i(qr).$$

Character multiplicativity and monotonicity in q force a power law.

One exponent per orientation determines all dot-product values:

$$\lambda(B_i(q)) = q^{p_i}, \qquad p_i \in [0,1], \quad i \in \{X, Y, Z\}.$$

Section 3. The three parameters need not be equal. Interpolation also gives monotonicity under degeneration. Spectral duality [2, 3] will turn character bounds into rank bounds.

Matrix multiplication factors into three dot products

In each term $a_{ik}b_{kj}c_{ij}$, the a, b, c variables belong to legs X, Y, Z, respectively.

Index	Appears in	Dot-product factor
Row $i \in [A]$	a_{ik}, c_{ij}	$B_Y(A)$
Inner $k \in [B]$	a_{ik}, b_{kj}	$B_Z(B)$
Column $j \in [C]$	b_{kj}, c_{ij}	$B_X(C)$

The subscript names the **scalar leg**. Tensoring pairs the indices:

$$\langle A, B, C\rangle \cong B_Y(A) \otimes B_Z(B) \otimes B_X(C).$$

$$\lambda(\langle A, B, C\rangle) = \lambda(B_Y(A))\lambda(B_Z(B))\lambda(B_X(C)) = A^{p_Y} B^{p_Z} C^{p_X}.$$

Sections 2–3. Pairing the factor coordinates gives precisely a_{ik}, b_{kj}, c_{ij}; coordinates on scalar legs have dimension one. The last line uses character multiplicativity and $\lambda(B_i(q)) = q^{p_i}$. Dimensions A, B, C are positive integers.

Shared sectors give a power-sum inequality

The pieces may share variables on leg i; their variables on the other two legs belong to disjoint, matched sectors.

Example: a shared first leg

$$T = \underbrace{xy_1z_1}_{T_1} + \underbrace{xy_2z_2}_{T_2}.$$

This is $B_X(2)$, so

$$\lambda(T) = 2^{p_X},$$
$$\lambda(T_1) = \lambda(T_2) = 1.$$

The general rule, for $p_i > 0$

If $T \trianglerighteq \sum_h T_h$ has this sector structure,

$$\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}.$$

The next slides prove this using tensor powers, finite separation, and counting.

Lemma 3.3. The common variable x makes the example a shared-leg sum. The following slides prove the rule using finite separation and multinomial counting. The separation construction is in the holdout slides.

Tensor powers group sector choices by multiplicity

Suppose $T \trianglerighteq \sum_{h=1}^{s} T_h$ has matched sectors sharing leg i. Fix a character λ with $p = p_i > 0$.

Expanding $(\sum_h T_h)^{\otimes N}$ gives a sector for every word $w_1 \cdots w_N \in \{1, \ldots, s\}^N$. For example, 112 means $T_1 \otimes T_1 \otimes T_2$:

$$\underbrace{112, \quad 121, \quad 211}_{\text{multiplicities } (n_1,n_2)=(2,1)} .$$

Fixing counts $n_1 + \cdots + n_s = N$ gives

$$M = \binom{N}{n_1, \ldots, n_s}, \qquad U = \bigotimes_{h=1}^{s} T_h^{\otimes n_h}.$$

Retain these M words by zeroing coordinates on the two unshared legs. Each word has the factors of U, in a different order.

Section 3, proof of Lemma 3.3. Here $N \geq 1$ and the $n_h \geq 0$ are integers. Independent reordering of the words' factors becomes valid after applying the separation lemma.

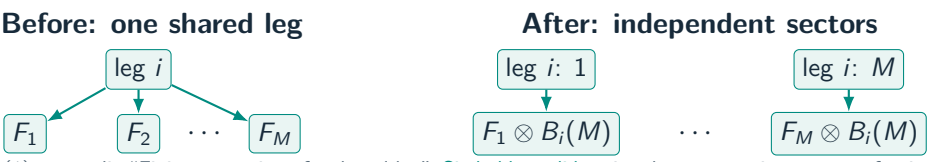
Finite separation: give every sector its own variables

Let $Q = F_1 + \cdots + F_M$, with $M \geq 1$ sectors sharing leg i. Their variables on each of the other two legs are pairwise disjoint.

Finite-separation lemma

$$(2M)Q \trianglerighteq \bigoplus_{w=1}^{M} (F_w \otimes B_i(M)).$$

Here $(2M)Q$ means $2M$ **independent copies** of Q.



Lemma 2.1(1); appendix "Finite separation of a shared leg". Six holdout slides give the construction: a root-of-unity filter of length $2M$, followed by ε-weights that retain matching sector labels.

Finite separation makes the words independent

Let $Q = \sum_{w=1}^{M} F_w$ be the retained word sectors. The previous retention step gives $T^{\otimes N} \trianglerighteq Q$; these sectors share leg i.

Apply the lemma, then reorder factors

$$(2M)\,T^{\otimes N} \trianglerighteq (2M)\,Q \trianglerighteq \bigoplus_{w=1}^{M} (F_w \otimes B_i(M)) \cong \bigoplus_{w=1}^{M} (U \otimes B_i(M)).$$

All three legs now distinguish w, so each word's factors can be reordered independently to give U.

First step: retain the multiplicity group in each copy. Second: the finite-separation lemma (Lemma 2.1(1)), proved in six holdout slides. Third: all words have the same factor multiplicities, so $F_w \cong U$.

A multiplicity group gives one character inequality

Apply λ to the separation degeneration, using monotonicity from interpolation. Since $p = p_i > 0$, we have $\lambda(B_i(M)) = M^p$:

$$2M\,\lambda(T)^N \geq \underbrace{M}_{\text{copies}} \underbrace{\prod_h \lambda(T_h)^{n_h}}_{\lambda(U)} \underbrace{M^p}_{\lambda(B_i(M))}.$$

Cancel M, take the power $1/p$, and set

$$a_h = \lambda(T_h)^{1/p}.$$

$$M \prod_{h=1}^{s} a_h^{n_h} \leq 2^{1/p} \lambda(T)^{N/p}.$$

The bound on the right is the same for every multiplicity tuple.

Lemma 3.3 and the appendix "Counting and rescaling". The calculation uses character additivity, multiplicativity, and monotonicity under degeneration; the latter follows from interpolation (Lemma 3.1; also shown in backup).

Sum over multiplicities, then take roots

There are at most $(N+1)^s$ tuples with $n_1 + \cdots + n_s = N$. Sum their bounds and use the multinomial theorem:

$$\left(\sum_{h=1}^{s} a_h\right)^N = \sum_{\substack{n_1+\cdots+n_s=N \\ n_h \geq 0}} \binom{N}{n_1, \ldots, n_s} \prod_h a_h^{n_h}$$
$$\leq (N+1)^s 2^{1/p} \lambda(T)^{N/p}.$$

Take N-th roots and let $N \to \infty$:

$$\sum_h a_h \leq \underbrace{(N+1)^{s/N} 2^{1/(pN)}}_{\longrightarrow 1} \lambda(T)^{1/p}.$$

This proves Lemma 3.3: $\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}$.

Lemma 3.3 and the appendix "Counting and rescaling". The number of sectors s and the character's parameter $p = p_i > 0$ stay fixed as $N \to \infty$.

Two-sector splitting lemma

Fix integers $r \geq 1$, $s \geq 2$, and distinct legs i, j.

Two matched sectors

$$C_{ij}(r,s) \otimes B_i(2) \trianglerighteq T_+ + T_-, \qquad T_\pm \cong C_{ij}(r, s \pm 1).$$

The sectors share leg i and use disjoint coordinates on the other legs.

Lemma 2.1(2); [1, Lemma 4.1]. Coordinate interpretation and proof in backup.

Three-sector splitting lemma

Fix integers $r, s \geq 1$ and distinct legs i, j, k.

Three matched sectors

$$C_{ij}(r, 3s + r - 1) \trianglerighteq T_1 + T_2 + T_3,$$
$$T_1, T_2 \cong C_{ij}(r, s), \qquad T_3 \cong C_{ik}(r, s).$$

The sectors share leg i and use pairwise disjoint coordinates on the other legs.

Lemma 2.1(3); [1, Lemma 4.2]. Coordinate interpretation and proof in backup.

The splittings give two character inequalities

Fix a character λ, put $p = p_i > 0$, and write $f_{ij}(r,s) := \lambda(C_{ij}(r,s))$.

Two sectors: neighboring lengths

Since $\lambda(B_i(2)) = 2^p$,

$$2f_{ij}(r,s)^{1/p} \geq f_{ij}(r,s+1)^{1/p} + f_{ij}(r,s-1)^{1/p}.$$

This is discrete concavity of $s \mapsto f_{ij}(r,s)^{1/p}$.

Three sectors: coupled orientations

$$f_{ij}(r,3s+r-1)^{1/p} \geq 2f_{ij}(r,s)^{1/p} + f_{ik}(r,s)^{1/p}.$$

The same parameter p_i controls both orientations, since leg i is shared.

Lemmas 2.1 and 3.3. Character multiplicativity evaluates the two-sector source; the power-sum rule evaluates the shared sectors. The first inequality requires $r \geq 1, s \geq 2$; the second $r, s \geq 1$.

Replication lets us compare all scales

Replication into m independent copies gives

$$C_{ij}(mr, ms) \trianglerighteq mC_{ij}(r, s) \quad \Longrightarrow \quad f_{ij}(mr, ms) \geq mf_{ij}(r, s).$$

Normalize the lengths; the limit exists for $r, s > 0$:

$$g_{ij}(r, s) = \lim_{m \to \infty} \frac{f_{ij}(\lceil mr \rceil, \lceil ms \rceil)}{m}.$$

For $p = p_i > 0$, powered concavity ($0 < \delta < s$) and tripling become

$$2g_{ij}(r, s)^{1/p} \geq g_{ij}(r, s - \delta)^{1/p} + g_{ij}(r, s + \delta)^{1/p},$$
$$g_{ij}(r, 3s + r)^{1/p} \geq 2g_{ij}(r, s)^{1/p} + g_{ik}(r, s)^{1/p}.$$

The additive -1 becomes $-1/m \to 0$ after rescaling.

The profiles are continuous and homogeneous: $g_{ij}(tr, ts) = t\, g_{ij}(r, s)$.

Lemmas 2.2 and 4.1; Appendices E–F. Polynomial division and evaluation give replication. Common-scale approximation preserves the inequalities; additional profile properties are in backup.

Concavity and tripling couple two profiles

Fix a leg i with $p_i > 0$; let j, k be the other legs. Write

$$P(s) := P_{ij}(s) = g_{ij}(1, s)^{1/p_i}, \qquad Q(s) := P_{ik}(s) = g_{ik}(1, s)^{1/p_i}.$$

Both profiles are positive and nondecreasing. Their inequalities are

Concavity, $0 < \delta < s$

$$2P(s) \geq P(s - \delta) + P(s + \delta),$$
$$2Q(s) \geq Q(s - \delta) + Q(s + \delta).$$

Tripling, $s > 0$

$$P(3s + 1) \geq 2P(s) + Q(s),$$
$$Q(3s + 1) \geq 2Q(s) + P(s).$$

We will prove that $\lim_{s\to\infty} \frac{P(s)}{s} = \lim_{s\to\infty} \frac{Q(s)}{s} = L_i > 0.$

Rescaled-profile inequalities (Lemma 4.1), with $r = 1$. The second tripling inequality exchanges j and k; the parameter p_i stays the same.

Tripling forces the limiting slopes to agree

The secant slopes $(P(s) - P(1))/(s-1)$ are nonnegative and decrease with $s > 1$. Thus their finite limit exists and equals

$$L_P := \lim_{s\to\infty} \frac{P(s) - P(1)}{s-1} = \lim_{s\to\infty} \frac{P(s)}{s} \geq 0.$$

Define L_Q analogously for Q.

Divide both tripling inequalities by s and let $s \to \infty$. Since $P(3s+1)/s \to 3L_P$, and likewise for Q,

$$\begin{aligned} 3L_P \geq 2L_P + L_Q \quad &\Longrightarrow \quad L_P \geq L_Q, \\ 3L_Q \geq 2L_Q + L_P \quad &\Longrightarrow \quad L_Q \geq L_P. \end{aligned}$$

$L_P = L_Q =: L_i \geq 0$. It remains to prove $L_i > 0$.

Proof of Lemma 5.1, Appendix G. Monotonicity makes the secants nonnegative; concavity makes them decrease.

Iterated tripling makes the common slope positive

Put $R = P + Q > 0$. Adding the tripling inequalities gives

$$R(3s+1) \geq 3R(s), \qquad \lim_{s\to\infty} \frac{R(s)}{s} = 2L_i.$$

Fix $s > 0$. Iterate at

$$s_0 = s, \quad s_{n+1} = 3s_n + 1 \quad \Longrightarrow \quad s_n = 3^n \left(s + \tfrac{1}{2}\right) - \tfrac{1}{2}.$$

Then $R(s_n) \geq 3^n R(s)$. Since $R(s_n)/s_n \to 2L_i$ and $s_n/3^n \to s + \frac{1}{2}$,

$$R(s) \leq \lim_{n\to\infty} \frac{R(s_n)}{3^n} = 2L_i \left(s + \tfrac{1}{2}\right) = L_i(2s+1).$$

$L_i \geq \dfrac{R(s)}{2s+1} > 0$: the common slope is positive.

Proof of Lemma 5.1, Appendix G. The same estimate $P(s) + Q(s) \leq L_i(2s+1)$ will bound the offsets.

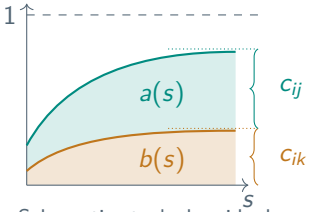
The two residuals share a budget of one

Divide $P(s) + Q(s) \le L_i(2s+1)$ by $L_i > 0$ and subtract $2s$:

$$\underbrace{\left(\frac{P(s)}{L_i} - s\right)}_{a(s)} + \underbrace{\left(\frac{Q(s)}{L_i} - s\right)}_{b(s)} \le 1.$$

Concavity makes a, b nonnegative and nondecreasing. The budget bounds each by 1, so both limits exist:

$$c_{ij} := \lim_{s\to\infty} a(s),$$
$$c_{ik} := \lim_{s\to\infty} b(s).$$



Schematic stacked residuals

$c_{ij}, c_{ik} \ge 0, \quad c_{ij} + c_{ik} \le 1.$
$L_i s \le P(s) \le L_i(s + c_{ij}), \quad L_i s \le Q(s) \le L_i(s + c_{ik}).$

Lemma 5.1, Appendix G. The limits exist because the residuals are nondecreasing and bounded; $a(s) \le c_{ij}$ and $b(s) \le c_{ik}$.

The profile bounds force relative growth

Derivative sketch. Assume $p_i, p_j > 0$ and differentiable $g = g_{ij}$.

$$P'_{ij}(t) \geq L_i, \quad P_{ij}(t) \leq L_i(t + c_{ij}) \quad \Longrightarrow \quad \frac{P'_{ij}(t)}{P_{ij}(t)} \geq \frac{1}{t + c_{ij}}.$$

Homogeneity gives $g(r, s) = r\, P_{ij}(s/r)^{p_i}$; differentiate in s:

$$\frac{\partial_s g}{g} = \frac{p_i}{r} \frac{P'_{ij}(s/r)}{P_{ij}(s/r)} \geq \frac{p_i/r}{s/r + c_{ij}} = \frac{p_i}{s + c_{ij} r}.$$

Exchange inputs and apply the same bounds with i, j swapped. With $P_{ji}(t) = g_{ji}(1, t)^{1/p_j}$,

$$g(r, s) = g_{ji}(s, r) = s\, P_{ji}(r/s)^{p_j} \quad \Longrightarrow \quad \frac{\partial_r g}{g} \geq \frac{p_j}{r + c_{ji} s}.$$

Lemma 5.1 and Proposition 5.2, Appendix G. Concavity gives $P'_{ij} \geq L_i$; the affine bound controls the denominator. The finite-increment proof is in backup.

The two relative-growth bounds share a budget

Derivative sketch. For differentiable $g = g_{ij}$, differentiate $g(tr, ts) = t\, g(r, s)$ at $t = 1$:

$$r\, \partial_r g + s\, \partial_s g = g \quad \Longrightarrow \quad r\frac{\partial_r g}{g} + s\frac{\partial_s g}{g} = 1.$$

Insert the two derivative lower bounds from the preceding slide:

$$1 \geq \frac{p_j r}{r + c_{ji} s} + \frac{p_i s}{s + c_{ij} r} \qquad (r, s > 0).$$

Optimize over r/s to obtain a constraint between p_i, p_j and their offsets.

Proposition 5.2 and Appendix G. The rigorous proof derives the same inequality using finite increments, as shown in backup. This growth argument treats positive parameters.

A quadratic bounds each pair of offsets

Set $t = r/s > 0$ and clear the positive denominators in the preceding inequality:

$$\underbrace{(1-p_j)c_{ij}}_{A}\,t^2 + \underbrace{\bigl(1-p_i-p_j+c_{ij}c_{ji}\bigr)}_{B}\,t + \underbrace{(1-p_i)c_{ji}}_{C} \geq 0.$$

Here $A, C \geq 0$, and the inequality holds for every $t > 0$. Divide by t and optimize:

$$B \geq -\inf_{t>0}(At + C/t) = -2\sqrt{AC}.$$

Substitution and completing the square give

$$\left(\sqrt{c_{ij}c_{ji}} + \sqrt{(1-p_i)(1-p_j)}\right)^2 \geq p_i p_j.$$

$$\sqrt{c_{ij}c_{ji}} \geq \sqrt{p_i p_j} - \sqrt{(1-p_i)(1-p_j)}.$$

Proposition 5.2, Appendix G. We use $0 \leq p_i \leq 1$ and $c_{ij} \geq 0$. The infimum formula also covers zero A or C.

The optimization problem before substitution

Fix weights $a, b, c \geq 0$. The numerical problem is

$$\begin{aligned} \text{maximize} \quad & W(p) := ap_1 + bp_2 + cp_3 \\ \text{subject to} \quad & 0 \leq p_i \leq 1, \\ & c_{ij} \geq 0, \qquad c_{ij} + c_{ik} \leq 1, \\ & \sqrt{c_{ij}c_{ji}} \geq \sqrt{p_i p_j} - \sqrt{(1-p_i)(1-p_j)}. \end{aligned}$$

The constraints hold for all distinct $i, j, k \in \{1, 2, 3\}$.

We seek an upper bound on W for every feasible tuple. Next write $p_i = \cos^2 \theta_i$.

The variables are the three parameters and six offsets; the weights are fixed. These numerical constraints are the only hypotheses used below.

The optimization problem in the angles

Set $p_i = \cos^2 \theta_i$. For fixed weights $a, b, c \geq 0$, the problem is

$$\begin{aligned} \text{maximize} \quad & W(\theta) := a \cos^2 \theta_1 + b \cos^2 \theta_2 + c \cos^2 \theta_3 \\ \text{subject to} \quad & 0 \leq \theta_i \leq \pi/2, \\ & c_{ij} \geq 0, \qquad c_{ij} + c_{ik} \leq 1, \\ & \sqrt{c_{ij} c_{ji}} \geq \cos(\theta_i + \theta_j). \end{aligned}$$

The constraints hold for all distinct $i, j, k \in \{1, 2, 3\}$.

First eliminate the offsets to obtain a constraint on the angles. Then bound the objective W.

The variables are the three angles and six offsets; the weights are fixed. These are exactly the numerical constraints after the substitution $p_i = \cos^2 \theta_i$.

The constraints force the angle-sum bound

Goal: prove $S := \theta_1 + \theta_2 + \theta_3 \geq \dfrac{\pi}{2}$.

If some $\theta_i \geq \pi/2$, we are done; otherwise all $\theta_i < \pi/2$.

Multiply each pairwise bound by $2\cos\theta_i \cos\theta_j$ and use AM–GM:

$$2\cos(\theta_i + \theta_j)\cos\theta_i \cos\theta_j \leq c_{ij}\cos^2\theta_i + c_{ji}\cos^2\theta_j.$$

Sum over pairs, use $c_{ij} + c_{ik} \leq 1$, and factor:

$$\begin{aligned} 0 &\geq 2\sum_{i<j} \cos(\theta_i + \theta_j)\cos\theta_i\cos\theta_j - \sum_i \cos^2\theta_i \\ &= \cos S \underbrace{\sum_i \cos\theta_i \cos(\theta_j - \theta_k)}_{>0} \quad \Longrightarrow \quad S \geq \pi/2. \end{aligned}$$

This proves Proposition 5.2 using the constraints alone. The factorization follows from cosine addition. In the last sum, j, k are the other indices; each cosine is positive since $\theta_i < \pi/2$ and $|\theta_j - \theta_k| < \pi/2$.

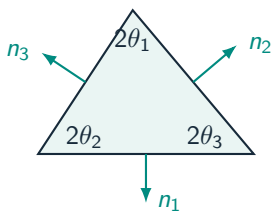
The angular boundary gives triangle geometry

Since scaling down the angles only increases $W(\theta)$, we may assume $\sum_i \theta_i = \pi/2$ for an upper bound. We relax the original constraints to maximizing $W(\theta) = \sum_i a_i \cos^2 \theta_i$ over all nonnegative angles with this sum, where $(a_1, a_2, a_3) = (a, b, c)$.

Take a triangle with angles $2\theta_i$.

Let n_i be the outward unit normal to the side opposite $2\theta_i$. The angle between n_j, n_k is $\pi - 2\theta_i$, so

$$n_j \cdot n_k = -\cos(2\theta_i).$$



For any original tuple of sum $S \geq \pi/2$, scaling by $\pi/(2S)$ reaches this boundary with an objective at least as large. Degenerate triangles follow by continuity.

A squared norm bounds the weighted objective

For $a, b, c > 0$, set

$$\ell_1 = \sqrt{\frac{bc}{a}}, \qquad \ell_2 = \sqrt{\frac{ca}{b}}, \qquad \ell_3 = \sqrt{\frac{ab}{c}}.$$

Their pairwise products are the weights: $\ell_2\ell_3 = a$, $\ell_3\ell_1 = b$, $\ell_1\ell_2 = c$. Set $v = \ell_1 n_1 + \ell_2 n_2 + \ell_3 n_3$. Using the normal-vector identity,

$$\|v\|^2 = \sum\nolimits_i \ell_i^2 - 2\sum\nolimits_i a_i \cos(2\theta_i).$$

Since $\cos^2\theta_i = (1 + \cos(2\theta_i))/2$, define

$$E = E(a, b, c) := \frac{a+b+c}{2} + \frac{1}{4}\left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}\right).$$

$$W = E - \frac{1}{4}\|v\|^2 \le E.$$

An elementary proof of the Wolstenholme step [4] in Section 6. All three angles enter the same squared-norm identity.

The thin case gives $W \le a + b$

Assume the **thin case**: $a \ge b \ge c > 0$ and $c \le \frac{ab}{a+b}$. Then

$$\ell_3 - \ell_1 - \ell_2 = \frac{ab - ac - bc}{\sqrt{abc}} \ge 0.$$

Since each n_i is a unit vector, the triangle inequality gives

$$\|v\| \ge \ell_3 - \|\ell_1 n_1 + \ell_2 n_2\| \ge \ell_3 - \ell_1 - \ell_2.$$

Insert this lower bound into $W = E - \|v\|^2/4$:

$$W \le E - \frac{(\ell_3 - \ell_1 - \ell_2)^2}{4} = E - \frac{(ab - ac - bc)^2}{4abc} = a + b.$$

The case $c = 0$ follows directly from $0 \le p_i \le 1$. Equality at the threshold $c = ab/(a + b)$ is included.

Character bounds give arithmetic algorithms

Published spectral input:

$$\widetilde{R}(T) = \max_{\lambda} \lambda(T), \qquad \widetilde{R}(T) = \inf_{N \geq 1} \operatorname{rank}(T^{\otimes N})^{1/N}.$$

If every character has $ap_Y + bp_Z + cp_X \leq E$:

1. Choose a fixed large base B and round the block dimensions upward.
2. Spectral duality supplies a finite exact rank identity for a tensor power.
3. Apply that identity recursively to matrix blocks; pad to the input size.

For every positive exponent slack, we obtain a fixed finite block algorithm. Recursion gives the claimed arithmetic exponent.

Theorem 3.2 [2, 3] and Lemma 6.1. Linear combinations cost $O(\ell R^{\ell})$ over ℓ recursion levels. The exponent statement does not provide a practical small-base implementation.

Applications through established tradeoffs

Problem	Running time
Directed unweighted APSP: Zwick's balance [7]	$O(n^{2.5})$
Combining with Alman–Vassilevska Williams [8]	$O(n^{2.4999})$
One LCA per pair in a DAG [9]	$O(n^{(3+\sqrt{3})/2})$
Exact all-pairs Hamming distance [10, 11]	$O(n^{(13-\sqrt{7})/4})$
Minimum / maximum Boolean witnesses [12]	$O(n^{2.5})$
Sparse multiplication, $m \leq n^{1.25}$ [13]	$O(n^2)$

Section 7. The LCA and Hamming exponents are approximately 2.36603 and 2.58856, respectively. Hamming inputs are n vectors of length n over an unrestricted alphabet; sparse inputs have at most m nonzeros each.

What the rectangular extension contributes

Starting constructions

The two polynomial splittings recap [1]. Finite separation adapts its shared-leg construction.

Rectangular analysis

Keep all orientations; replicate and rescale convolution profiles; couple six offsets; optimize the weighted angular constraint.

Standard tools

Multinomial counting, spectral duality, interpolation, squared norms, and the triangle inequality.

Lean correspondence

The main bounds are checked with the published spectral theorem supplied explicitly as a hypothesis.

$$c \le \frac{ab}{a+b} \implies \omega(a,b,c) = a+b \qquad (a \ge b \ge c \ge 0).$$

The note makes no priority claim for standard tools or finite construction variants. Lean checks mathematical statements; some finite proof routes differ from the manuscript. The application algorithms are outside the formalization.

Backup: the degeneration target has rank three

The coefficients of c_0 and c_1 in $S = a_0 b_0 c_0 + (a_0 b_1 + a_1 b_0) c_1$ are the two matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A rank-two decomposition of S would put both A and B in the span of two rank-one matrices. Since A, B are independent, those matrices would have to span exactly the same plane.

$$\alpha A + \beta B = \begin{pmatrix} \alpha & \beta \\ \beta & 0 \end{pmatrix}, \qquad \det(\alpha A + \beta B) = -\beta^2.$$

Every nonzero rank-one matrix in this plane is therefore a multiple of A. Two independent rank-one generators cannot exist.

Thus $\operatorname{rank}(S) > 2$. Its three displayed monomials give $\operatorname{rank}(S) \leq 3$, so $\operatorname{rank}(S) = 3$.

Elementary slice-space proof over $\mathbb{C}$; the same rank statement is in [6, proof of Proposition 13, p. 131]. In a rank-two decomposition, the two bilinear factors are rank-one matrices; both output coefficients are linear combinations of them. Return to the example.

Separation proof: start with a shared first leg

Work over $\mathbb{C}$. Assume leg X is shared, and write

$$Q = \sum_{h=1}^{M} F_h, \qquad F_h = \sum_{a,b,c} \mu_{habc}\, x_a y_{h,b} z_{h,c}.$$

The x_a may be shared across sectors. The $y_{h,b}$ and $z_{h,c}$ are already private to sector h.

The target has independent variables on every leg

$$S := \bigoplus_{h=1}^{M} (F_h \otimes B_X(M)).$$

We construct an exact restriction, then a degeneration:

$$(2M)Q \;\geq\; R \;\trianglerighteq\; S.$$

Proof of Lemma 2.1(1), appendix "Finite separation of a shared leg". Here $M \geq 1$; all coordinate sets are finite. Other shared-leg orientations follow by permuting the three legs.

Mix $2M$ copies using roots of unity

Set $L = 2M$, and choose a primitive L-th root of unity ζ. Label the independent copies by $\ell = 0, \ldots, L-1$.

In copy ℓ, make these linear substitutions:

$$x_a^{(\ell)} \longmapsto \sum_{g=1}^{M} \zeta^{\ell g} X_{a,g},$$

$$y_{h,b}^{(\ell)} \longmapsto \sum_{u=1}^{M} \zeta^{\ell(u-h)} Y_{h,b,u},$$

$$z_{h,c}^{(\ell)} \longmapsto \frac{1}{L} \sum_{v=1}^{M} \zeta^{-\ell v} Z_{h,c,v}.$$

The new indices g, u, v each run from 1 to M. Each substitution acts on just one leg.

All new coordinates are independent within each leg. The first-leg label is g; u, v label coordinates on the other legs. Summing copies cancels unwanted combinations of these labels.

The finite sum keeps exactly $g + u = h + v$

After summing the copies, the coefficient of $X_{a,g}Y_{h,b,u}Z_{h,c,v}$ is μ_{habc} times

$$\frac{1}{L}\sum_{\ell=0}^{L-1}\zeta^{\ell(g+u-h-v)} = \begin{cases} 1, & g+u-h-v \equiv 0 \pmod{L}, \\ 0, & \text{otherwise.} \end{cases}$$

Since $|g + u - h - v| \le 2M - 2 < L$, the only multiple of L in this range is zero. Thus the filter keeps precisely $g + u = h + v$.

The exact restriction obtained so far

$$(2M)Q \ge R := \sum_{\substack{h,g,u,v \\ g+u=h+v}} \sum_{a,b,c} \mu_{habc} X_{a,g} Y_{h,b,u} Z_{h,c,v}.$$

The filter identity is a finite geometric sum: if $\zeta^d \neq 1$, then $\sum_{\ell=0}^{L-1}(\zeta^d)^\ell = 0$; otherwise the sum is L. All h, g, u, v lie in $\{1, \dots, M\}$.

Choose weights whose sum is a square

Every monomial of R satisfies $g + u = h + v$. For $\varepsilon \neq 0$, scale its coordinates by

$$\begin{aligned} X_{a,g} &\longmapsto \varepsilon^{g^2} X_{a,g}, \\ Y_{h,b,u} &\longmapsto \varepsilon^{2hu-h^2} Y_{h,b,u}, \\ Z_{h,c,v} &\longmapsto \varepsilon^{-2hv} Z_{h,c,v}. \end{aligned}$$

A surviving monomial acquires the total exponent

$$\begin{aligned} g^2 + 2hu - h^2 - 2hv &= g^2 + 2h(u - v) - h^2 \\ &= g^2 + 2h(h - g) - h^2 = (g - h)^2 \geq 0. \end{aligned}$$

The exponent is zero exactly when $g = h$; all other retained monomials have positive exponent.

Each weight depends only on indices of the coordinate being scaled, so the three substitutions act independently on the legs. The relation $u - v = h - g$ comes from the preceding filter.

Take the limit: matching labels survive

After scaling, the resulting tensor is

$$P(\varepsilon) = \sum_{\substack{h,g,u,v \\ g+u=h+v}} \sum_{a,b,c} \mu_{habc}\, \varepsilon^{(g-h)^2} X_{a,g}\, Y_{h,b,u} Z_{h,c,v}.$$

As $\varepsilon \to 0$, only $g = h$ survives. The filter equation then forces $u = v$:

$$\lim_{\varepsilon \to 0} P(\varepsilon) = \sum_{h=1}^{M} \sum_{a,b,c} \sum_{u=1}^{M} \mu_{habc} X_{a,h} Y_{h,b,u} Z_{h,c,u} =: S.$$

For $\varepsilon \neq 0$, $R \geq P(\varepsilon)$. The coefficientwise limit proves $R \trianglerighteq S$, hence $(2M)Q \trianglerighteq S$.

Negative powers are allowed in the substitutions for nonzero ε. After the finite filter, every nonzero coefficient has the nonnegative exponent $(g-h)^2$, so the limit exists.

Identify the independent sectors and dot products

Write $S = \sum_{h=1}^{M} S_h$, where

$$S_h = \sum_{a,b,c} \sum_{u=1}^{M} \mu_{habc} X_{a,h} Y_{h,b,u} Z_{h,c,u}.$$

Recall $B_X(M) = s \sum_{u=1}^{M} t_u w_u$. For each fixed h, pair the coordinates in $F_h \otimes B_X(M)$:

F_h coordinate	$B_X(M)$ coordinate	Paired coordinate
x_a	s	$X_{a,h}$
$y_{h,b}$	t_u	$Y_{h,b,u}$
$z_{h,c}$	w_u	$Z_{h,c,u}$

Thus $S_h \cong F_h \otimes B_X(M)$. All three legs carry h, so different S_h use disjoint variables.

$$(2M)Q \trianglerighteq S \cong \bigoplus_{h=1}^{M} (F_h \otimes B_X(M)).$$

This completes the proof of Lemma 2.1(1). Permuting the legs gives the same construction for any shared leg i.

Two-sector proof: the source and target maps

Fix $r \geq 1, s \geq 2$. Let $\mathcal{P}_d = \{p(t) : \deg p < d\}$ and $f \in \mathcal{P}_r$.

Source: lengths s, s

$$(f, (a, b)) \longmapsto (fa, fb),$$
$$(a, b) \in \mathcal{P}_s \oplus \mathcal{P}_s.$$

Target: lengths $s+1, s-1$

$$(f, (p, h)) \longmapsto (fp, fh),$$
$$(p, h) \in \mathcal{P}_{s+1} \oplus \mathcal{P}_{s-1}.$$

The first input f is shared. The second input contains two independent polynomials; the two output blocks are disjoint.

Lemma 2.1(2), appendix "Splitting two polynomial products". Return to the lemma statement.

Why tensoring with $B_X(2)$ gives two products

Recall: $B_X(2) = (uv_0)w_0 + (uv_1)w_1$ encodes outputs (uv_0, uv_1).

Read X, Y as inputs and Z as output. Both factors are bilinear maps:

$$C_{XY}(r,s) : \mathcal{P}_r \times \mathcal{P}_s \to \mathcal{P}_{r+s-1}, \qquad (f,g) \longmapsto fg,$$
$$B_X(2) : \mathbb{C} \times \mathbb{C}^2 \to \mathbb{C}^2, \qquad (u,(v_0,v_1)) \longmapsto (uv_0, uv_1).$$

The tensor product pairs corresponding input and output spaces:

$$\underbrace{(\mathcal{P}_r \otimes \mathbb{C})}_{\mathcal{P}_r} \times \underbrace{(\mathcal{P}_s \otimes \mathbb{C}^2)}_{\mathcal{P}_s \oplus \mathcal{P}_s} \longrightarrow \underbrace{(\mathcal{P}_{r+s-1} \otimes \mathbb{C}^2)}_{\mathcal{P}_{r+s-1} \oplus \mathcal{P}_{r+s-1}}.$$

For the standard basis e_0, e_1 of $\mathbb{C}^2$, this map sends

$$(f \otimes 1,\ a \otimes e_0 + b \otimes e_1) \longmapsto fa \otimes e_0 + fb \otimes e_1.$$

Thus $(f,(a,b)) \mapsto (fa, fb)$: two products sharing f.

The parameters $r, s, 2$ specify dimensions, not function arguments.

Two-sector proof: encode the inputs

Fix arbitrary $r \geq 1, s \geq 2$. The inputs are $f \in \mathcal{P}_r$ and $(p, h) \in \mathcal{P}_{s+1} \oplus \mathcal{P}_{s-1}$. Keep f unchanged.

Write $p_s = [t^s]p$. For $\varepsilon \neq 0$, encode the second input by

$$E_\varepsilon : (p, h) \longmapsto (a, b), \qquad \begin{cases} a = p - p_s t^s - \varepsilon^{-1} th, \\ b = \varepsilon^{-1} h + p_s t^{s-1}. \end{cases}$$

Both have degree at most $s - 1$: the leading term of p was removed, and $\deg(th) \leq s - 1$.

Apply the source bilinear map:

$$(f, (p, h)) \xrightarrow{\mathrm{id} \times E_\varepsilon} (f, (a, b)) \longmapsto (U, V) = (fa, fb).$$

Lemma 2.1(2), appendix "Splitting two polynomial products". E_ε is a linear map from $\mathcal{P}_{s+1} \oplus \mathcal{P}_{s-1}$ to $\mathcal{P}_s \oplus \mathcal{P}_s$, independently of f. The definitions give $a + tb = p$.

Two-sector proof: decode the outputs

Put $m = r + s - 2$, so $\deg U, \deg V \le m$. Decode the output pair linearly:

$$D_\varepsilon : (U, V) \longmapsto (P, H_\varepsilon), \qquad \begin{cases} P = U + tV, \\ H_\varepsilon = \varepsilon(V - [t^m]V\, t^m). \end{cases}$$

The output lengths are $m + 2 = r + s$ for P and $m = r + s - 2$ for H_ε.

Since $a + tb = p$,

$$P = U + tV = f(a + tb) = fp.$$

Expand the other product:

$$V = \varepsilon^{-1} fh + p_s t^{s-1} f.$$

Because $\deg(fh) \le r + s - 3 = m - 1$, its top coefficient is

$$[t^m]V = p_s f_{r-1}, \qquad f_{r-1} = [t^{r-1}]f.$$

Coefficient extraction, deleting a coefficient, and shifting polynomial coefficients are linear output operations. The identity $[t^m]V = p_s f_{r-1}$ is used to analyze that output map.

Two-sector proof: the error vanishes

Substitute the expression for V and its top coefficient:

$$\begin{aligned} H_\varepsilon &= \varepsilon\big(V - p_s f_{r-1} t^{r+s-2}\big) \\ &= fh + \varepsilon p_s t^{s-1}\big(f - f_{r-1} t^{r-1}\big) \xrightarrow{\varepsilon \to 0} fh. \end{aligned}$$

Encoding the second input, applying the source map, and decoding gives

$$(f, (p, h)) \longmapsto (fp,\ \ fh + \varepsilon p_s t^{s-1}(f - f_{r-1} t^{r-1})).$$

The coefficients converge to those of (fp, fh). The independent inputs p, h have lengths $s+1, s-1$, and the two output blocks are disjoint; both products share f.

$$C_{XY}(r,s) \otimes B_X(2) \trianglerighteq T_+ + T_-, \qquad T_\pm \cong C_{XY}(r, s \pm 1).$$

This proves Lemma 2.1(2) for arbitrary $r \geq 1, s \geq 2$. If $r = 1$, the error is identically zero. Permuting the legs gives every orientation i, j.

Three-sector proof: inputs and outputs

Fix $r, s \geq 1$ and use $i = X, j = Y, k = Z$. Write $\mathcal{P}_d = \{p(t) : \deg p < d\}$.

The source map is $(f, g) \mapsto fg$, with $f \in \mathcal{P}_r$ and $g \in \mathcal{P}_{3s+r-1}$.

The sectors share f; their Y, Z coordinate blocks are disjoint:

Sector	Leg j	Leg k
T_1, T_2	Input: s coordinates	Output: $r + s - 1$ coordinates
T_3	Output: $r + s - 1$ coordinates	Input: s coordinates

Lemma 2.1(3), appendix "The three-sector splitting". The middle sector's first-leg reversal identifies that individual tensor. Return to the lemma statement.

Three sectors, with X, Y kept as the input legs

Split the second input into independent coefficient blocks:

$$g = g_L + t^s q + t^{2s+r-1} g_R, \qquad g_L, g_R \in \mathcal{P}_s, \quad q \in \mathcal{P}_{r+s-1}.$$

The target of the degeneration is the bilinear map

$$(f, (g_L, q, g_R)) \longmapsto \underbrace{(fg_L,\ M_f(q),\ fg_R)}_{\text{three output blocks}}, \qquad f \in \mathcal{P}_r,$$

where the middle output keeps s coefficients of fq:

$$M_f(q) := \sum_{v=0}^{s-1} [t^{r-1+v}](fq)\, t^v \in \mathcal{P}_s.$$

The outer outputs fg_L, fg_R each lie in $\mathcal{P}_{r+s-1}$.

There are two input groups: f and (g_L, q, g_R). The name C_{XZ} for the middle sector reads Z as input and Y as output; the proof below makes this identification explicit.

Here $[t^k]$ extracts the coefficient of t^k. The same three-leg tensor can be read with different choices of output leg.

Three-sector proof: partition the coordinates

Let x_a, y_b be coefficients of f, g, and z_c output labels. Set $n = 3s + r - 1$:

$$C_{XY}(r, n) = \sum_{a=0}^{r-1} \sum_{b=0}^{n-1} x_a y_b z_{a+b}.$$

Partition the integer indices of Y and Z into these disjoint intervals:

	Left	Middle	Right
Y	$[0, s-1]$	$[s, 2s+r-2]$	$[2s+r-1, n-1]$
Z	$[0, s+r-2]$	$[s+r-1, 2s+r-2]$	$[2s+r-1, n+r-2]$

Encode the second input as $g_\varepsilon = g_L + \varepsilon t^s q + t^{2s+r-1} g_R$. Decode by dividing the middle output block by ε; leave the outer output blocks unchanged.

These are diagonal linear maps on the second input and the output. The first input f is unchanged.

Lemma 2.1(3), appendix "The three-sector splitting". All intervals include both endpoints; the Y range is $0, \ldots, n-1$ and the Z range is $0, \ldots, n+r-2$.

Three-sector proof: only matched blocks survive

A nonzero term has $c = a + b$ with $0 \le a \le r - 1$.

$$b \le s - 1 \implies c \le s + r - 2 \quad \text{(left } Y \text{ implies left } Z\text{)},$$
$$b \ge 2s + r - 1 \implies c \ge 2s + r - 1 \quad \text{(right } Y \text{ implies right } Z\text{)}.$$

The multipliers on the possible blocks are therefore

$Y \backslash Z$	Left	Middle	Right
Left	1	0	0
Middle	ε	1	ε
Right	0	0	1

No supported term has a negative power of ε. As $\varepsilon \to 0$, only the three matched blocks remain.

$C_{XY}(r, n) \trianglerighteq T_1 + T_2 + T_3$, where T_1, T_2 are the left and right matched blocks, and T_3 is the middle matched block.

Zero means no supported terms. An ε-block vanishes (and may be empty); the middle–middle factor is $\varepsilon\varepsilon^{-1} = 1$.

Three-sector proof: identify the outer blocks

The left block has $0 \le b < s$. Every $a + b$ lies in the left Z interval, so

$$T_1 = \sum_{a=0}^{r-1} \sum_{v=0}^{s-1} x_a y_v z_{a+v} = C_{XY}(r, s).$$

Put $A = 2s + r - 1$. In the right block, write $b = A + v$, with $0 \le v < s$:

$$T_2 = \sum_{a=0}^{r-1} \sum_{v=0}^{s-1} x_a y_{A+v} z_{A+a+v}.$$

Relabel $y'_v = y_{A+v}$ and $z'_d = z_{A+d}$. Then

$$T_2 = \sum_{a=0}^{r-1} \sum_{v=0}^{s-1} x_a y'_v z'_{a+v} \cong C_{XY}(r, s).$$

Both outer tensors are copies of $C_{XY}(r, s)$. The first-leg basis is the same in both; their Y, Z coordinates are disjoint.

The displayed index changes are bijections on the corresponding coordinate intervals. The full ranges follow from the partition and $0 \le a < r$, $0 \le v < s$.

Three-sector proof: identify the middle block

Write $c = s + r - 1 + v$ and $a = r - 1 - u$, with $0 \le v < s$, $0 \le u < r$. Then $b = c - a = s + u + v \in [s, 2s + r - 2]$. These choices cover exactly the middle block:

$$T_3 = \sum_{u=0}^{r-1} \sum_{v=0}^{s-1} x_{r-1-u} y_{s+u+v} z_{s+r-1+v}.$$

Put $q_d = y_{s+d}$ and $h_v = z_{s+r-1+v}$. Reading X, Z as inputs, define their polynomials by

$$f^{\leftarrow}(t) = \sum_{u=0}^{r-1} x_{r-1-u} t^u, \qquad h(t) = \sum_{v=0}^{s-1} h_v t^v.$$

The coefficient of the output label q_d is

$$\sum_{u+v=d} x_{r-1-u} h_v = [t^d](f^{\leftarrow} h).$$

Thus $(f, h) \mapsto f^{\leftarrow} h$, giving $T_3 \cong C_{XZ}(r, s)$ by first-leg reversal.

Reversal identifies this sector individually; the degeneration keeps the common X coordinates. This proves Lemma 2.1(3); permuting the legs gives every orientation.

Backup: size bounds and input interchange

Size and monotonicity

For all $r, s > 0$,

$$\min\{r, s\} \le g_{ij}(r, s) \le r + s.$$

The profile is nondecreasing in each length.

Interchanging inputs gives first-coordinate concavity

$$g_{ij}(r, s) = g_{ji}(s, r).$$

For fixed $s > 0$ and $p_j > 0$, the right-hand profile is powered-concave in its second argument. Thus

$$r \longmapsto g_{ij}(r, s)^{1/p_j} \quad \text{is concave.}$$

Lemma 4.1 and Appendix F. The power for varying the first length is $1/p_j$, obtained by exchanging the two polynomial inputs.

Backup: why degenerations preserve character bounds

For a degeneration $T \trianglerighteq S$, interpolation (slide 20) gives

$$(NL+1)\,T^{\otimes N} \geq S^{\otimes N}.$$

Apply character additivity, multiplicativity, and restriction monotonicity:

$$\lambda(S)^N \leq (NL+1)\lambda(T)^N.$$

Take N-th roots:

$$\lambda(S) \leq \underbrace{(NL+1)^{1/N}}_{\longrightarrow 1}\lambda(T).$$

Let $N \to \infty$: $T \trianglerighteq S$ implies $\lambda(S) \leq \lambda(T)$.

Lemma 3.1 and Appendix D; [6, Theorem 8]. Here L is the degree of the transformed polynomial tensor in the interpolation construction.

Backup: the growth inequality without differentiability

The slope and affine bounds imply, for positive r, s, t,

$$\frac{g_{ij}(r, s+t)}{g_{ij}(r, s)} \geq \left(1 + \frac{t}{s + c_{ij} r}\right)^{p_i}.$$

Increase r by εr, then s by εs:

$$\begin{aligned} 1 + \varepsilon &= \frac{g_{ij}((1+\varepsilon)r, (1+\varepsilon)s)}{g_{ij}(r, s)} \\ &\geq \left(1 + \frac{\varepsilon r}{r + c_{ji} s}\right)^{p_j} \left(1 + \frac{\varepsilon s}{s + c_{ij}(1+\varepsilon) r}\right)^{p_i}. \end{aligned}$$

Take logarithms, divide by ε, and let $\varepsilon \downarrow 0$:

$$1 \geq \frac{p_j r}{r + c_{ji} s} + \frac{p_i s}{s + c_{ij} r}.$$

Appendix G, equation (G.2) and its consequence. Only the explicit elementary lower bound is differentiated; the profile itself is not.

Backup: formalization and provenance

Proof routes and scope.

Pinned finite splitting certificates and local-polynomial replication differ from the manuscript's exact constructions. The formalization also checks field descent; the slides follow the note's complex-field presentation. Applications are not formalized.

The accompanying note, its proofs, analysis and writeup were obtained fully through OpenAI Codex (GPT-6.1). The author has read and analyzed the note in full, takes full responsibility for its contents, and takes no credit for this work. These slides are a preliminary presentation of the note.

Repository: `lean/README.md` and `lean/PROOF-MAP.md`. Source theorem input: [2, 3].

References: proof ingredients

[1] OpenAI. *An upper bound of* 9/4 *for the matrix multiplication exponent* (2026). October 2 preprint.

[2] V. Strassen. *The asymptotic spectrum of tensors* (1988). DOI: 10.1515/crll.1988.384.102.

[3] A. Wigderson and J. Zuiddam. *Asymptotic spectra: theory, applications and extensions* (2026). DOI: 10.1090/bull/1880.

[4] J. Liu. *An improved result of a weighted trigonometric inequality in acute triangles with applications* (2020). DOI: 10.7153/jmi-2020-14-11.

[5] C. Ju and E. Solomonik. *Derivation and analysis of fast bilinear algorithms for convolution* (2020). DOI: 10.1137/19M1301059.

[6] M. Christandl, A. K. Jensen and J. Zuiddam. *Tensor rank is not multiplicative under the tensor product* (2018). DOI: 10.1016/j.laa.2017.12.020.

References: algorithmic tradeoffs

[7] U. Zwick. *All pairs shortest paths using bridging sets and rectangular matrix multiplication* (2002). DOI: 10.1145/567112.567114.

[8] J. Alman and V. Vassilevska Williams. *Truly subquadratic 3SUM and truly subcubic APSP via triangles in sparse lopsided graphs* (2026). DOI: 10.48550/arXiv.2610.06783.

[9] F. Grandoni, G. F. Italiano, A. Łukasiewicz, N. Parotsidis and P. Uznański. *All-pairs LCA in DAGs: breaking through the $O(n^{2.5})$ barrier* (2021). DOI: 10.1137/1.9781611976465.18.

[10] K. Labib, P. Uznański and D. Wolleb-Graf. *Hamming distance completeness* (2019). DOI: 10.4230/LIPIcs.CPM.2019.14.

[11] R. Yuster. *Efficient algorithms on sets of permutations, dominance, and real-weighted APSP* (2009). DOI: 10.1137/1.9781611973068.103.

[12] K. Cohen and R. Yuster. *On minimum witnesses for Boolean matrix multiplication* (2014). DOI: 10.1007/s00453-012-9742-3.

[13] R. Yuster and U. Zwick. *Fast sparse matrix multiplication* (2005). DOI: 10.1145/1077464.1077466.